

6.3 Eurodollar Futures

The most popular interest rate futures contract in the United States is the **three-month Eurodollar futures contract** traded on the CME Group.

A **Eurodollar** is:

- A US dollar deposit
- Held in a bank outside the United States

The Eurodollar interest rate is essentially the same as:

LIBOR (London Interbank Offered Rate)

It represents the interest rate at which banks lend US dollars to one another.

Nature of the Eurodollar Futures Contract

A three-month Eurodollar futures contract is:

A futures contract on the interest rate that will apply to a \$1 million deposit for a future three-month period.

The contract allows traders to:

- Speculate on future short-term interest rates
 - Hedge future borrowing/lending costs
 - Construct the LIBOR zero curve
 - Manage money-market exposure
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Contract Characteristics

Underlying

- Interest on \$1 million for 3 months

Contract Months

Available maturities:

- March
- June
- September
- December

Contracts may extend:

- Up to 10 years into the future

Thus traders can take positions on future interest rates many years ahead.

Final Settlement

The final settlement occurs:

- Two business days before the third Wednesday of the delivery month

At settlement:

$$\text{Final Settlement Price} = 100 - R$$

Where:

- R = 3-month LIBOR fixing
 - Expressed with:
 - Quarterly compounding
 - Actual/360 day count convention
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Example — Final Settlement

Suppose:

- Final LIBOR fixing = 0.75%

Then:

$$100 - 0.75 = 99.250$$

Thus:

Final settlement price = 99.250

After final settlement:

- All contracts are closed

Price Quotation Convention

Eurodollar futures are quoted as:

$$100 - \text{Interest Rate}$$

This means:

Futures Quote	Implied Interest Rate
99.725	0.275%
97.400	2.600%
96.500	3.500%

Important Intuition

Because the quote is:

there is an inverse relationship between:

- Interest rates
 - Futures prices
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Relationship Between Futures Price and Interest Rates

If Interest Rates Rise

- Futures prices fall

If Interest Rates Fall

- Futures prices rise

Therefore:

Position	Profits When
Long Futures	Rates fall
Short Futures	Rates rise

One Basis Point Rule

A very important feature of Eurodollar futures:

A one-basis-point move equals \$25 per contract

Why?

A basis point:

$$0.01\% = 0.0001$$

Contract principal:

$$\$1,000,000$$

Three-month period:

$$0.25 \text{ years}$$

Interest change:

$$1,000,000 \times 0.0001 \times 0.25 = 25$$

Thus:

1 basis point = \$25

Trading Profit and Loss

If Quote Increases by 1 Basis Point

- Long gains \$25
- Short loses \$25

If Quote Decreases by 1 Basis Point

- Long loses \$25
 - Short gains \$25
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Example — Daily Gain/Loss

Suppose settlement price changes:

$$99.725 \rightarrow 99.685$$

Difference:

$$0.040 = 4 \text{ basis points}$$

Loss to long position:

$$4 \times 25 = \$100$$

Gain to short position:

$$\$100$$

Contract Price Formula

Hull defines the contract price as:

$$10,000 [100 - 0.25(100 - Q)]$$

Where:

- Q = futures quote
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Example — Contract Price

Suppose:

$$Q = 99.725$$

Then:

$$\begin{aligned} &10,000 [100 - 0.25(100 - 99.725)] \\ &= 10,000 [100 - 0.25(0.275)] \\ &= 10,000(99.93125) \end{aligned}$$

$$= 999,312.5$$

Thus:

Contract Price = \$999,312.50

Example — Final Contract Value

Suppose final quote becomes:

$$99.615$$

Then:

$$\begin{aligned} 10,000 [100 - 0.25(100 - 99.615)] \\ = 999,037.5 \end{aligned}$$

Difference:

$$999,312.5 - 999,037.5 = 275$$

Thus:

Long position loses \$275

Hedging with Eurodollar Futures

Eurodollar futures are widely used for:

- Hedging future investments
 - Locking future borrowing rates
 - Locking future lending rates
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Example 6.3 — Locking in an Interest Rate

Suppose an investor wants to lock in the interest rate for:

- A future 3-month investment
- Principal = \$100 million

Current September Eurodollar futures quote:

96.50

Implied interest rate:

$$100 - 96.50 = 3.5\%$$

The investor buys:

100 contracts

Suppose Actual Future Rate Becomes 2.6%

Final settlement price:

$$100 - 2.6 = 97.40$$

Price movement:

$$97.40 - 96.50 = 0.90 = 90 \text{ basis points}$$

Profit on Futures Position

Each basis point:

\$25

Therefore:

$$100 \times 25 \times 90 \\ = 225,000$$

Thus:

Futures profit = \$225,000

Interest Earned on Investment

Actual investment return:

$$100,000,000 \times 0.25 \times 0.026 \\ = 650,000$$

Total Effective Return

$$650,000 + 225,000 = 875,000$$

Equivalent interest rate:

$$100,000,000 \times 0.25 \times 0.035 = 875,000$$

Thus:

Effective locked-in rate = 3.5%

Important Reality — Hedge Is Not Perfect

Hull explains that the hedge is not exactly perfect because:

1. Futures Are Settled Daily

Cash flows occur every day via marking to market.

2. Timing Mismatch Exists

- Futures settle in September
- Actual investment interest received in December

This creates reinvestment risk.

Hedge Adjustment

Hull suggests an approximate adjustment:

$$\frac{1}{1 + 0.035 \times 0.25} = 0.9913$$

Adjusted contracts:

$$100 \times 0.9913 = 99$$

Thus:

Use 99 contracts instead of 100

Eurodollar Futures and the Yield Curve

Eurodollar futures provide information about:

- Future LIBOR expectations
- Interest rate term structure

Example from Hull:

Future Date	Futures Rate
June 2013	0.275%
September 2013	0.295%
December 2013	0.325%
December 2015	0.900%
December 2017	2.270%
December 2019	3.324%

This indicates:

An upward-sloping yield curve

Meaning:

- Long-term interest rates are higher than short-term rates.
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Forward Rates vs Futures Rates

Eurodollar futures resemble:

Forward Rate Agreements (FRAs)

because both lock in future interest rates.

Important Difference Between FRAs and Futures

Eurodollar Futures

- Settled daily
- Final settlement at time T_1

FRA

- No daily settlement
 - Settlement occurs at time T_2
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Two Major Differences

1. Daily Settlement Effect

Daily settlement creates cash flows during the contract life.

This changes contract value because:

- Gains occur when rates move favorably
- Margin cash can earn interest

This creates:

Convexity Bias

2. Timing of Settlement

Futures:

- Settled earlier

FRAs:

- Payoff occurs later

Because of time value of money:

- Futures rates tend to exceed forward rates.
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Convexity Adjustment

Hull gives the convexity adjustment formula:

$$\text{Forward Rate} = \text{Futures Rate} - \frac{1}{2}\sigma^2 T_1 T_2$$

Where:

- σ = volatility of short-term rates
 - T_1 = maturity of futures contract
 - T_2 = maturity of underlying rate
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Key Insight

Convexity adjustment:

- Reduces futures rate
 - Produces estimated forward rate
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Example 6.4 — Convexity Adjustment

Suppose:

$$\sigma = 0.012$$

8-year Eurodollar futures quote:

Implied futures rate:

$$6\%$$

Given:

$$T_1 = 8$$

$$T_2 = 8.25$$

Calculate Adjustment

$$\begin{aligned} \frac{1}{2}(0.012)^2(8)(8.25) \\ = 0.00475 \end{aligned}$$

Thus:

$$0.475\% = 47.5 \text{ basis points}$$

Forward Rate Estimate

Continuous-compounding futures rate:

$$6.038\%$$

Adjusted forward rate:

$$\begin{aligned} 6.038 - 0.475 \\ = 5.563\% \end{aligned}$$

Thus:

$$\text{Forward rate} = 5.563\%$$

Important Observation

Hull notes:

Convexity adjustment grows rapidly with maturity

Maturity	Adjustment
2 years	3.2 bps
4 years	12.2 bps
6 years	27.0 bps
8 years	47.5 bps
10 years	73.8 bps

Observation:

- Adjustment approximately grows with:

$$T^2$$

Using Eurodollar Futures to Build the LIBOR Zero Curve

Eurodollar futures are frequently used to extend:

The LIBOR zero curve

Basic Idea

Known:

- Short-term LIBOR rates

Need:

- Longer-term zero rates

Eurodollar futures provide:

- Forward rates

These forward rates are bootstrapped into zero rates.

Relationship Between Forward and Zero Rates

Hull derives:

$$F_i = \frac{R_{i+1}T_{i+1} - R_iT_i}{T_{i+1} - T_i}$$

Rearranging:

$$R_{i+1} = \frac{F_i(T_{i+1} - T_i) + R_iT_i}{T_{i+1}}$$

Example 6.5 — Bootstrapping Zero Rates

Suppose:

400-day zero rate:

4.80%

Forward rate for next 91 days:

5.30%

Compute 491-Day Zero Rate

$$\frac{0.053 \times 91 + 0.048 \times 400}{491} \\ = 0.04893$$

Thus:

491-day zero rate = 4.893%

Compute 589-Day Zero Rate

Next forward rate:

$$5.50\%$$

Then:

$$\frac{0.055 \times 98 + 0.04893 \times 491}{589} \\ = 0.04994$$

Thus:

589-day zero rate = 4.994%

6.4 Duration-Based Hedging Strategies Using Futures

Duration is extremely important in interest rate risk management.

Suppose:

- A bond portfolio is being hedged
- Using interest rate futures

Define:

Symbol	Meaning
V_F	Futures contract value
D_F	Duration of underlying futures asset
P	Portfolio value
D_P	Portfolio duration

Duration Approximation

For small yield changes:

$$\Delta P = -PD_P \Delta y$$

Similarly:

$$\Delta V_F = -V_F D_F \Delta y$$

Duration-Based Hedge Ratio

Equating sensitivities gives:

$$N = \frac{PD_P}{V_F D_F}$$

Where:

- N = number of futures contracts

This is called:

Duration-Based Hedge Ratio

or

Price Sensitivity Hedge Ratio

Important Assumption

This approach assumes:

Parallel shifts in the yield curve

Meaning:

- All interest rates move equally

This is a major limitation.

Choosing the Correct Futures Contract

The hedger tries to choose futures contracts where:

$$D_F \approx D_P$$

Typical Usage

Futures Type	Usage
Eurodollar Futures	Short-term rates
Treasury Bond Futures	Long-term rates

Futures Type	Usage
Treasury Note Futures	Medium-term rates

Important Direction Rule

If Rates Rise

- Bond prices fall
- Futures prices fall

If Rates Fall

- Bond prices rise
- Futures prices rise

Hedging Rule

Exposure	Hedge
Lose when rates rise	Short futures
Lose when rates fall	Long futures

Example 6.6 — Duration Hedge

Suppose:

Portfolio value:

\$10,000,000

Portfolio duration:

6.80 years

Current futures price:

$$93 - 02 = 93.0625$$

Contract value:

93,062.50

CTD bond duration:

9.20 years

Calculate Contracts

$$\begin{aligned} N &= \frac{10,000,000 \times 6.80}{93,062.50 \times 9.20} \\ &= 79.42 \end{aligned}$$

Thus:

Short 79 futures contracts

Why Short Futures?

If rates rise:

- Bond portfolio loses value
- Short futures gains value

If rates fall:

- Bond portfolio gains value
- Short futures loses value

Thus hedge offsets interest-rate risk.

Cheapest-to-Deliver Risk

In Treasury bond futures:

- Multiple deliverable bonds exist

The hedger assumes one bond will be CTD.

If market conditions change:

- Another bond may become CTD

This weakens hedge effectiveness.

Business Snapshot — Asset Liability Management (ALM)

Banks carefully manage interest-rate exposure using:

Asset Liability Management (ALM)

GAP Management

Banks divide the yield curve into:

Buckets

Examples:

- 0–1 month
- 1–3 months
- 3–6 months

Banks analyze:

- How portfolio value changes when one bucket shifts.
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Tools Used by Banks

Banks use:

- Swaps
 - FRAs
 - Bond futures
 - Eurodollar futures
 - Other interest-rate derivatives
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6.5 Hedging Portfolios of Assets and Liabilities

Financial institutions often hedge by matching:

- Duration of assets
- Duration of liabilities

This is called:

Duration Matching

or

Portfolio Immunization

Objective of Immunization

Protect against:

Small parallel shifts in interest rates

How It Works

If:

$$\text{Asset Duration} = \text{Liability Duration}$$

then:

- Gain on assets offsets loss on liabilities
 - Portfolio becomes less sensitive to rates
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Limitation of Duration Matching

Duration matching only protects against:

Parallel yield curve shifts

In reality:

- Short-term rates are more volatile
- Long-term rates move differently

Sometimes:

- Short rates rise
- Long rates fall

Therefore:

Duration matching is only a first approximation

Summary of Chapter Concepts

Treasury Bond Futures

Important delivery options:

1. Delivery timing option
2. Choice of deliverable bond
3. Wild card option

These options:

Reduce futures price

Eurodollar Futures

- Based on future 3-month LIBOR
- Quoted as:

$$100 - R$$

- Used for:
 - Hedging
 - Speculation
 - Yield curve construction
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Convexity Adjustment

Necessary because:

- Futures $\neq$ forward contracts

Adjustment:

$$\text{Forward Rate} = \text{Futures Rate} - \frac{1}{2}\sigma^2 T_1 T_2$$

Duration Hedging

Hedge ratio:

$$N = \frac{PD_P}{V_F D_F}$$

Purpose:

- Protect against interest-rate changes
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Core Intuition of Entire Chapter

This chapter connects:

- Futures pricing
- Bond sensitivity
- LIBOR markets
- Hedging
- Yield curves
- Duration management

Key ideas:

1. Interest rates and bond prices move inversely.
2. Eurodollar futures dominate short-term interest-rate trading.
3. Treasury futures contain embedded delivery options.
4. Duration measures interest-rate sensitivity.

5. Hedging works best under parallel yield-curve shifts.
6. Convexity adjustment is necessary when using futures to estimate forward rates.